

Literature Mining Report

Query Statistics

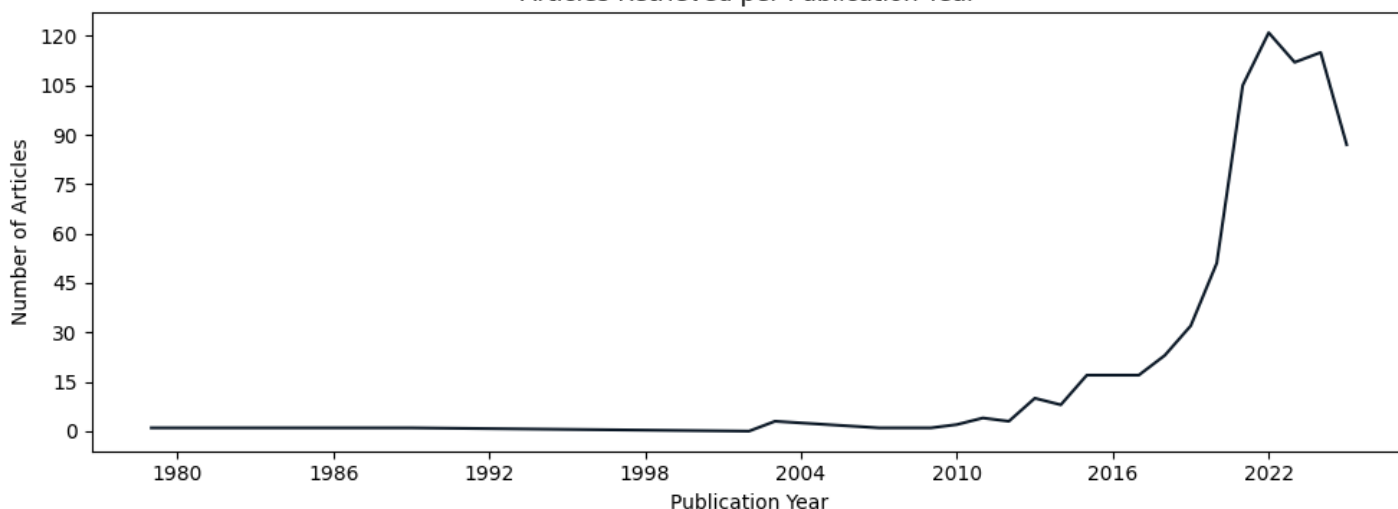
Query date: 2025-08-14

Query terms: "microbial induced concrete calcite precipitation"

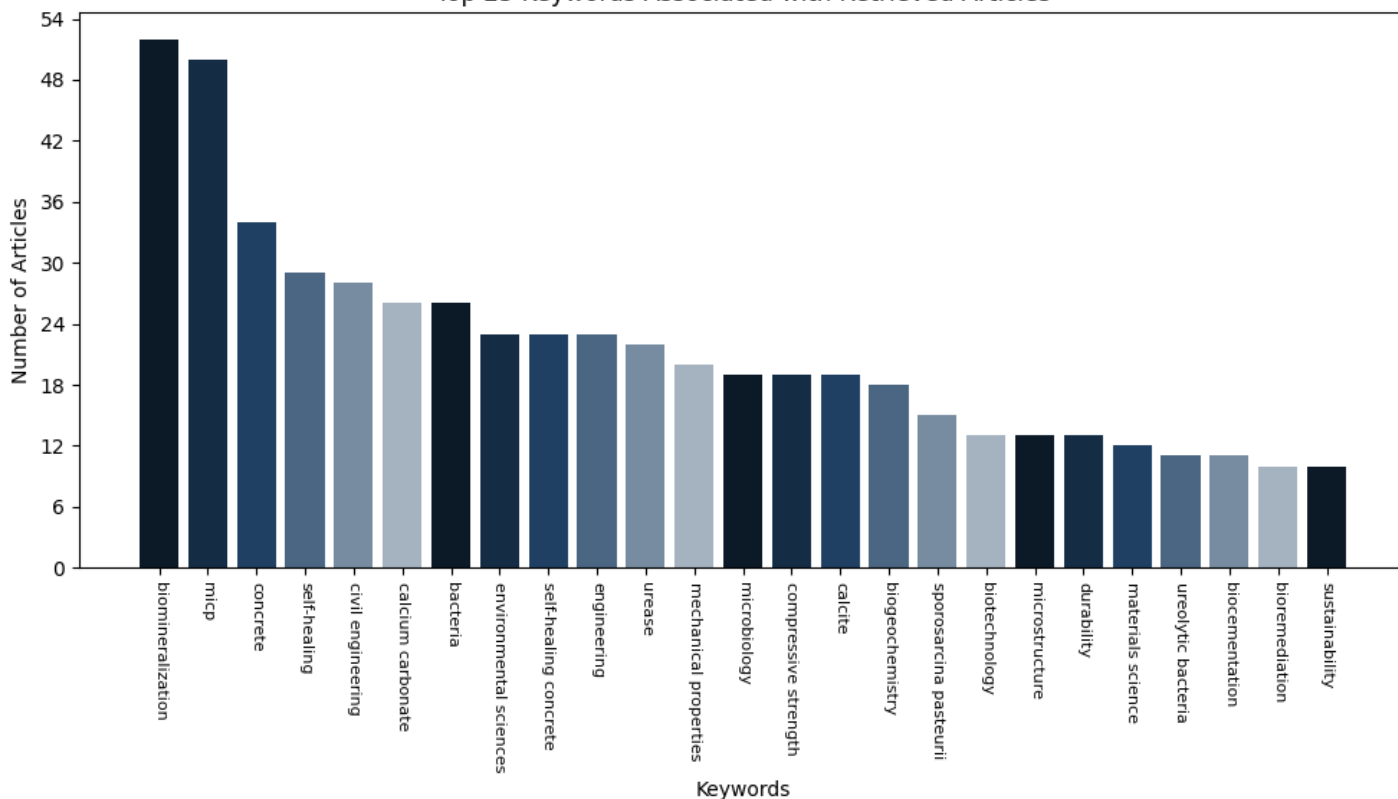
Articles parsed/hits: 752/752 (100.00%)

Scoring keywords: concrete, calcite, precipitation

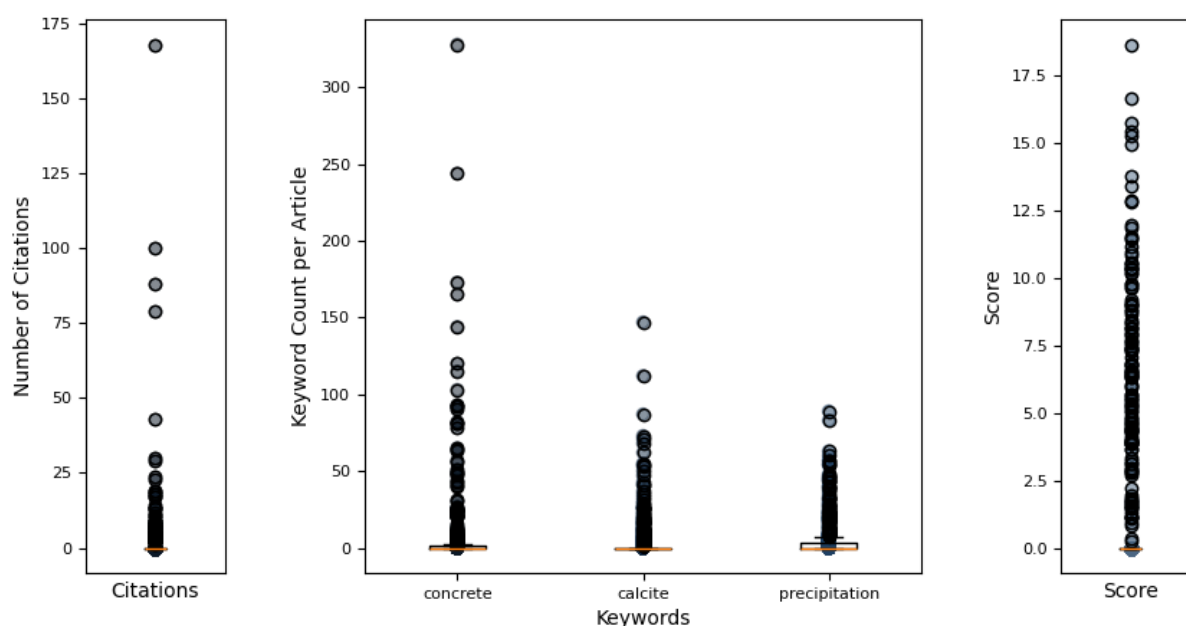
Articles Retrieved per Publication Year



Top 25 Keywords Associated with Retrieved Articles



Hermes Report Summary



Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
38972424	2024	The Science of	Wong, Pui Yan et al.	advances in microbial self-healing concrete: ...	7	18.66
40246949	2025	Scientific Repo	Shivaprakash, S et al.	effect of natural carbonates on microbially i...	0	16.67
39673695	2025	Journal of Indu	Elgendy, Ibrahi et al.	bacteria-powered self-healing concrete: break...	4	15.76
31980427	2020	Applied and Env	Reeksting, Bian et al.	in-depth profiling of calcite precipitation b...	24	15.43
33561160	2021	PLoS ONE	Zehner, Jennife et al.	calcite seed-assisted microbial induced carbo...	7	15.28
34319256	2021	Journal of Micr	Han, Pei-pei et al.	improvement of biomineralization of sporosarc...	6	14.95
33629805	2021	Microbial Biote	Shaheen, Nafees et al.	isolation of alkaliphilic calcifying bacteria...	0	13.80
34794517	2021	Fungal Biology	Van Wylick, Aur et al.	a review on the potential of filamentous fung...	0	13.43
26835451	2016	Frontiers in Bi	Zhu, Tingting et al.	carbonate precipitation through microbial act...	0	12.85
40595441	2025	Scientific Repo	B, Sunantha et al.	a comprehensive screening on the effect of ba...	0	12.83
37128499	2023	Environmental E	Zhang, Kuan et al.	microbialinduced carbonate precipitation (mi...	0	11.96
40542343	2025	ACS Applied Mat	Tuyishime, J. R et al.	nanoscale, bchar: bterization of fungal...	0	11.85
36224231	2022	Scientific Repo	Burdalski, Robe et al.	mineralogy, morphology, and reaction kinetics...	0	11.56
40374753	2025	Scientific Repo	Wang, Jianxin et al.	evaluation and optimal width ratio selection ...	0	11.50
39628580	2024	iScience	Jung, Patrick et al.	hard shell, soft blue-green core: ecology, pr...	0	11.42
40594560	2025	Scientific Repo	Danial, Amal W. et al.	assessment of ecofriendly carbon capture usin...	0	11.17
34211000	2021	Scientific Repo	Zehner, Jennife et al.	a sample cell for the study of enzyme-induced...	0	10.98
40328859	2025	Scientific Repo	Vedrtnam, Ajita et al.	finite element simulation of bacterial self-h...	0	10.83
35855981	2022	Heliyon	Nasser, Amal A. et al.	microbially-induced-calcite-precipitation (mi...	0	10.60
32317706	2020	Scientific Repo	Huang, Yi-Hsun et al.	efficient option of industrial wastewater res...	0	10.55
36576609	2022	World Journal o	Erdmann, N. et al.	influencing factors on ureolytic microbiologi...	11	10.53
33409575	2021	AMB Express	Hong, Minyoung et al.	agricultural by-products and oyster shell as ...	0	10.39
35517729	2020	RSC Advances	Zehner, Jennife et al.	microbial-induced calcium carbonate precipita...	0	10.36
39323773	2024	Heliyon	Rasheed, Sajid et al.	development of biological techniques to preve...	0	10.35
24194735	2013	Frontiers in Mi	Dhami, Navdeep et al.	biomineralization of calcium carbonates and t...	0	10.31

Query Result 1/25

Advances in microbial self-healing concrete: A critical review of mechanisms, developments, and future directions

Wong, Pui Yan; Mal, Joyabrata; Sandak, Anna; Luo, Lijun; Jian, Jianxiong; Pradhan, Nirakar

The Science of the Total Environment

Year: 2024 PMID: 38972424 doi: 10.1016/j.scitotenv.2024.174553 Citations: 7 Score: 18.66

Keywords hits: concrete: 244, calcite: 13, precipitation: 55

Abstract:

The self-healing bioconcrete, or bioconcrete as concrete containing microorganisms with self-healing capacities, presents a transformative strategy to extend the service life of concrete structures. This technology harnesses the biological capabilities of specific microorganisms, such as bacteria and fungi, which are integral to the materials capacity to autonomously mend cracks, thereby maintaining structural integrity. This review highlights the complex biochemical pathways these organisms utilize to produce healing compounds like calcium carbonate, and how environmental parameters, such as pH, temperature, oxygen, and moisture critically affect the repair efficacy. A comprehensive analysis of recently published peer-reviewed literature, and contemporary experimental research forms the backbone of this review with a focus on microbiological aspects of the self-healing process. The review assesses the challenges facing self-healing bioconcrete, including the longevity of microbial spores and the cost implications for large-scale implementation. Further, attention is given to potential research directions, such as investigating alternative biological agents and optimizing the concrete environment to support microbial activity. The culmination of this investigation is a call to action for integrating self-healing bioconcrete in construction on a broader scale, thereby realizing its potential to fortify infrastructure resilience and sustainability.

Summary:

The ability of bacteria to create an alkaline environment is considered their primary function in this process. In the urea-based MICCP approach, urease plays a crucial role in hydrolysing urea and producing ammonia and CO₂, leading to an elevated pH within the ureolytic bacteria cell. Additionally, due to the negatively charged surfaces of the bacterial cell and the increased concentration of CO₂ within the crack, the bacterial cell serves as a nucleation site for the heterogeneous precipitation of CaCO₃ crystals. Ureolytic bacteria pivotal to the MICCP process are typically heterotrophic facultatives that thrive under alkaline conditions where their metabolic functions increase pH, aiding in the precipitation of CaCO₃ and also ureolytic bacteria, including halophiles and alkaliphiles can produce bioconcrete at pH levels ranging from 8.5 to 11. For instance, *Sporosarcina pasteurii*, an ureolytic bacteria, hydrolyzes urea best at pH 9.25 and cease to grow beyond pH 1. On the other hand, fungi typically found in aerobic environments, can also survive extreme conditions by forming spores, suggesting potential for their use in industrial self-healing bioconcrete applications. Introducing algae species into concrete fractures could be a future study to assess their feasibility in crack healing and contribute to more sustainable concrete solutions. Apart from exploring novel microorganisms for bioconcrete application, it is also important to have in-depth knowledge of the viability and metabolisms of microorganisms under various environmental conditions for enhanced self-healing performance of the microorganisms as extreme environmental factors can have a serious detrimental impact on the long-term self-healing efficiency.

Mentioned biomedical entities:

Genes: N2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Chrysosporium, Culture Collection Ex.)4Factors, E. coli, MICCP of D. radiodurans, Penicillium, Penicillium chrysogenum, Purpureocillium lilacinum spores, Trichoderma longibrachiatum, bacterial cell, bacterial cells, calcium.(Li, cell, cells, cellular, cereus, halodurans spores, layer, material, spore, subtilis, tablet

Organisms: Bacillus cereus, Bacillus cereus CS1, E. coli, human, humans, lung, skin

Tissues: crack zone, mineral, smooth

Pathways: N/A

Query Result 2/25

Effect of natural carbonates on microbially induced calcite precipitation process

Shivaprakash, Shaivan H.; Yanez, Valerie R.; Graddy, Charles M. R.; Gomez, Michael G.; DeJong, Jason T.; Burns, Susan E.
Scientific Reports

Year: 2025 PMID: 40246949 doi: 10.1038/s41598-025-97737-2 Citations: 0 Score: 16.67

Keywords hits: concrete: 48, calcite: 147, precipitation: 4

Abstract:

Microbially induced calcite precipitation (MICP) is an emerging ground improvement technique that uses microbes to induce cementation between soil particles. To date, the majority of research has focused on exploring MICP with silica-rich sands; however, the present study investigates the process and efficacy of MICP in a carbonate-rich natural soil, and a comparison is made with benchmark silica-rich sands. MICP column experiments were performed with a range of treatment formulations to optimize and understand the MICP process in carbonate-rich soil. Performance was quantified using chemical (pH, urea, and ammonium concentrations) and physical measurements (TGA and LOI tests). Micro-scale characterization of the cemented soils was performed with XRD, SEM, and EDS, while shear-wave velocity (Vs) and unconfined compressive strength tests were performed to evaluate the effect of precipitated calcite on macroscopic engineering properties. Natural carbonates were found to have a significant impact on the MICP process, resulting in an increase in MICP efficiency of 23% and increases in precipitated calcite contents by as much as 82% when compared to benchmark silica-rich soils receiving similar treatments. These results suggest that the presence of natural carbonate minerals within soils may lower the energy barrier and act as preferential sites for calcite precipitation during the MICP process. Furthermore, SEM images highlighted the association of bacterial cells with precipitated calcite crystals, differences in calcite morphologies and more widespread cementation bonds in carbonate-rich soil when compared to silica sand. Generated cementation also resulted in a linear increase in Vs with increases in precipitated calcite contents for MICP treated carbonate-rich soil, consistent with past results for silica sands. Lastly, differences in yeast extract concentrations applied in treatment solutions were also found to significantly impact the development of ureolytic microbial capacity and the efficiency of the MICP process in the considered soils.

Summary:

For the same flow rate conditions, increasing yeast extract concentration from .2 g/L to 2 g/L resulted in a high increase in MICP efficiency calculated as /Mass of input Ca²⁺ in cementation solution) in carbonate-rich Blessington soil from 48 to 72% compared to 44% to 49% in silica-rich F-11 sand and, the increase in calcite precipitation was as high as 8% in Blessington sand compared to 27% in Ottawa F-11 sand. The precipitated calcite contents of bio-cemented samples also agree well with the TGA results and the precipitated calcite profiles from acid digestion tests reported earlier. Table 6 Loss on Ignition Results for Untreated and MICP Treated Blessington, Concrete, and Ottawa F-11 Sand Samples. Sample Loss in mass Measured calcite content Precipitated calcite content Untreated Blessington sand 13.263.16 Untreated Concrete sand 2.93 Untreated F-11 sand 2.251.116.236.446.28 BS.2351.4:115.6635.615.46 BS.251:115.8636.65.91 BS.2351:115.8135.965.8 BS.251:1COM15.535.255.9 CS.251:1COM3.68.188.18 CS.251:13.357.637.63 OS.2351:11.73.873.87 Microscale tests X-ray diffraction X-ray diffraction was performed on soil samples to obtain their mineralogical composition, and to confirm the presence of the precipitated calcite mineral phase in the MICP treated soil samples. 13 Shear-wave velocity profiles with increasing cementation treatments for different MICP treatment formulations of Blessington and concrete sands. Comparing Vs profiles between carbonate-rich soil and silica-rich sands for the same treatment formulations: for yeast extract concentration of .2 g/L, the increase in calcite precipitation in Blessington soil was .6% compared to F-11 sand, which was not sufficient to increase the stiffness of the Blessington column to equal/greater than Ottawa sand column. However, when the yeast extract concentration was increased to 2 g/L, the increase in calcite precipitation in Blessington soil was 4% compared to F-11 sand, which resulted in a steady increase in Vs of the Blessington soil column, achieving Vs values similar to F-11 sand column towards the end of cementation treatment and equaling it at 1th cementation cycle, despite containing 3% fines, because a higher amount of calcite content would be needed to establish cementation bonds between Blessington soil particles as compared to F-11 sand which contains sand particles of approx. For the same flow rate conditions, increasing yeast extract concentration from .2 g/L to 2 g/L resulted in a high increase in MICP efficiency in carbonate-rich Blessington soil from 48 to 72% compared to 44% to 49% in silica-rich F-11 sand, and the increase in calcite precipitation was as high as 8% in Blessington sand compared to 27% in Ottawa F-11 sand, which shows the compounding effect of yeast extract concentration and natural carbonates on the MICP process. Shear-wave velocity measurements showed that the additional increase in precipitated calcite content observed in Blessington soil treated with yeast extract concentration of 2 g/L was responsible for stiffening of soil particle contact points due to formation of cementation bonds, thus leading to a steady increase in Vs values.

Mentioned biomedical entities:

Genes: IAP, MPa, S12, TG, d9

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: B) SEM, BS.251:1, BS.2351:1, F) 1th, MPa, SEM, b, carbonate-rich, cell, column, silica, ureolytic cell, scenddocumentinterfacial

Organisms: Leon, seeds

Tissues: Portland cement, bender element, mineral, specimens, specimens BS.251:1

Pathways: N/A

Query Result 3/25**Bacteria-powered self-healing concrete: Breakthroughs, challenges, and future prospects**

Elgendy, Ibrahim M; Elkaliny, Nehal E; Saleh, Hoda M; Darwish, Gehad O; Almostafa, Mervt M; Metwally, Kamel; Yahya, Galal; Mahmoud, Yehia A -G

Journal of Industrial Microbiology Biotechnology

Year: 2025

PMID: 39673695

doi: 10.1093/jimb/kuae051

Citations: 4

Score: 15.76

Keywords hits: concrete: 12, calcite: 1, precipitation: 25

Abstract:

In a world where concrete structures face constant degradation from environmental forces, a revolutionary solution has emerged: bio-self-healing concrete. This innovation involves embedding dormant bacteria within the concrete mix, poised to spring into action when cracks form. As moisture seeps into the cracks, these bacterial agents are activated, consuming nutrients and converting them into calcium carbonate, a natural substance that fills and repairs the fractures, restoring the materials integrity. This fascinating process represents a cutting-edge approach to maintaining concrete infrastructure, turning once-vulnerable materials into self-sustaining systems capable of healing themselves. The ongoing research into bio-self-healing concrete is focused on selecting bacterial strains that can withstand the extreme conditions within concrete, including its highly alkaline environment. The bacteria must also form resilient spores, remaining viable until they are needed for repair. Additionally, the study explores various challenges associated with this technology, such as the cost of production, the bacteria's long-term viability, and their potential environmental impact. Advancements in genetic engineering and smart technology are being explored to enhance these bacterial strains, making them more efficient and robust in their role as microscopic repair agents. This review delves into the potential of bio-self-healing concrete to revolutionize how we approach infrastructure maintenance, offering a glimpse into a future where concrete structures not only endure but actively repair themselves, extending their lifespan and reducing the need for costly repairs.

Summary:

This process is driven by a key chemical reaction where urea hydrolysis, catalyzed by the enzyme urease, results in the formation of carbonate ions that combine with calcium to produce the CaCO_3 needed for healing. Factors Affecting Biological Healing in Concrete Microbial population is defined as the number of microbes in concrete has an impact on its ability to heal biologically because urease productivity and CaCO_3 precipitation are both related to bacterial cell concentration. This concentration of cells results in the best concrete healing because it produces the most urease enzyme. Therefore, a temperature between 30C and 35C is optimal for biohealing, as it enhances urease activity and bacterial growth, leading to increased CaCO_3 production. Calcium Concentration The concentration of calcium ions is also a critical factor affecting the self-healing efficiency of bacteria in concrete because it impacts on bacterial calcium precipitation. Also, when bacteria are used to work for the healing of cracks in concrete, it must face the harsh environment of the concrete such as high alkalinity of concrete, restricting the growth of the bacteria, and limit the efficiency of bio-self-healing. This mutant strain was able to produce more EPS and biofilm than the wild-type strain. Genetic Engineering of the ure C Gene to Increase Urease Production The ure C gene encodes for the alpha subunit of urease, an enzyme that catalyzes the hydrolysis of urea into carbon dioxide and ammonia.

Mentioned biomedical entities:

Genes: EP, ureB, ureC, ureE

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Bacillus subtilis PTCC, C, E. coli, EC, Nanometer, S. pasteurii, bacterial cell, bacterial cells, cell, cell calcium, cells, cellular, material, plasmid, single-cell, specimens, spore, subtilis, ureolytic-type

Organisms: Bacillus megaterium, Bacillus pseudofirmus, Bacillus sphaericus, Bacillus subtilis, Escherichia coli, humans

Tissues: precracked cement, specimens

Pathways: N/A

Query Result 4/25**In-Depth Profiling of Calcite Precipitation by Environmental Bacteria Reveals Fundamental Mechanistic Differences with Relevance to Application**

Reeksting, Bianca J.; Hoffmann, Timothy D.; Tan, Linzhen; Paine, Kevin; Gebhard, Susanne; Kelly, Robert M.

Applied and Environmental Microbiology

Year: 2020

PMID: 31980427

doi: 10.1128/AEM.02739-19

Citations: 24

Score: 15.43

Keywords hits: concrete: 11, calcite: 18, precipitation: 54

Abstract:

Biomining triggered by bacteria is important in the natural environment and has many applications in industry and in civil and geotechnical engineering. The diversity in biomining capabilities of environmental bacteria is, however, not well understood. This study surveyed environmental bacteria for their ability to precipitate calcium carbonate minerals and investigated both the mechanisms and the resulting crystals. We show that while urease activity leads to the fastest precipitation, it is by no means essential. Importantly, the same quantities of calcium carbonate are produced by nonureolytic bacteria, and the resulting crystals appear to have larger volumes and more organic components, which are likely beneficial in specific applications. Testing both precipitation mechanisms in a self-healing concrete application showed that nonureolytic bacteria delivered more robust results. Here, we performed a systematic study of the fundamental differences in biomining between environmental bacteria, and we provide important information for the design of bacterially based engineering solutions.

Summary:

The precipitation of calcium carbonate crystals throughout the medium was likely due to a change in the saturation kinetics in the medium, resulting from the rapid increase in pH caused by the urease-mediated release of ammonia from urea. These results show that the differences in ureolytic activity across our environmental isolates not only affected the kinetics of biomining but also had a clear impact on the crystal morphologies in the resulting precipitate. Profound effects of urease activity on precipitation of calcium carbonate. To exclude the possibility that the observed differences in calcium carbonate precipitation were simply due to strain-to-strain variation among our isolates, we exploited the fact that CG73 was capable of switching between ureolytic and nonureolytic states depending on the availability of urea (Fig. Mortars containing either ureolytic or nonureolytic bacteria also displayed crack healing, and this generally extended across the top, as well as down the sides, of the crack (Fig. In contrast to initial expectations that this mechanism may lead to the formation of small amounts of precipitate, we found that all of our isolates were capable of precipitating the total amount of calcium supplied in the growth media, although this process took more time with nonureolytic strains. Test broth was inoculated with 1 l of an overnight culture grown in LB and incubated at 37C for up to 5 days.

Mentioned biomedical entities:

Genes: 16S rRNA

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: B) cells, C, CEM, Cells, Fig. S8, LB, METHODS Bacterial isolates, Material, S. pasteurii, bacterial cells, cell, cells, colonies, cultures, high-pH, layer, nonureolytic bacteria, nonureolytic isolates, nonureolytic strains, phase-bright, solid medium, spore

Organisms: Bacillus licheniformis, Bacillus muralis, Bacillus pseudofirmus, Human

Tissues: Deutsche Sammlung von Mikroorganismen und Zellkulturen, layer, specimens

Pathways: N/A

Query Result 5/25

Calcite seed-assisted microbial induced carbonate precipitation (MICP)

Zehner, Jennifer; Ryne, Anja; Sikorski, Pawel; Mukherjee, Amitava

PLoS ONE

Year: 2021

PMID: 33561160

doi: 10.1371/journal.pone.0240763

Citations: 7

Score: 15.28

Keywords hits: concrete: 5, calcite: 112, precipitation: 48

Abstract:

Microbial-induced calcium carbonate precipitation (MICP) is a biological process inducing biomineralization of CaCO_3 . This can be used to form a solid, concrete-like material. To be able to use MICP successfully to produce solid materials, it is important to understand the formation process of the material in detail. It is well known that crystallization surfaces can influence the precipitation process. Therefore, we present in this contribution a systematic study investigating the influence of calcite seeds on the MICP process. We focus on the changes in the pH and changes of the optical density (OD) signal measured with absorption spectroscopy to analyze the precipitation process. Furthermore, optical microscopy was used to visualize the precipitation processes in the sample and connect them to changes in the pH and OD. We show, that there is a significant difference in the pH evolution between samples with and without calcite seeds present and that the shape of the pH evolution and the changes in OD can give detailed information about the mineral precipitation and transformations. In the presented experiments we show, that amorphous calcium carbonate (ACC) can also precipitate in the presence of initial calcite seeds and this can have implications for consolidated MICP materials.

Summary:

Experiments were performed with four different bacteria cell concentrations, as specified in Table 1.1.1371/journal.pone.24763. g1 Fig 1 Real time evolution of pH and OD75nm for MICP experiments without the presence of calcite seeds for the first 55 min of the reaction. The increase in OD75nm was observed for both samples only during the pH increase in the initial phase of the reaction. The precipitation process for the sample D2-C2 was investigated with optical microscopy to correlate changes in the pH and the OD75nm to the precipitation process. The scale-bar is 1 μm . Dried crystals for unseeded and seeded samples were characterized with SEM.1.1371/journal.pone.24763. g4 Fig 4 SEM characterization of precipitated crystals for unseeded and seeded samples for high bacterial concentration and lower bacterial cell concentration. Furthermore, initial calcite seeds have a different surface compared to the MICP induced calcite, as can be seen in. Discussion In the presented experiments, we investigated the influence of calcite seeds on the CaCO_3 precipitation in MICP processes. We observe significant differences in the pH evolution for unseeded and seeded samples. With SEM characterization, we could show that the precipitated CaCO_3 connected the initial seeds together. Furthermore, it was observed for unseeded samples that the bacterial cell concentration had a strong influence on the size and shape of the precipitated calcite crystals.

Mentioned biomedical entities:

Genes: SACC

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, b) 1, b) Bottom, bacterial cell, bacterial cells, cell, cell concentration.ACC, cells, layer, seeds

Organisms: blue arrow, seeds

Tissues: layer, seed, seeds

Pathways: N/A

Query Result 6/25**Improvement of Biomineralization of *Sporosarcina pasteurii* as Biocementing Material for Concrete Repair by Atmospheric and Room Temperature Plasma Mutagenesis and Response Surface Methodology**

Han, Pei-pei; Geng, Wen-ji; Li, Meng-nan; Jia, Shi-ru; Yin, Ji-long; Xue, Run-ze

Journal of Microbiology and Biotechnology

Year: 2021

PMID: 34319256

doi: 10.4014/jmb.2104.04019

Citations: 6

Score: 14.95

Keywords hits: concrete: 26, calcite: 47, precipitation: 2

Abstract:

Microbially induced calcium carbonate precipitation (MICP) has recently become an intelligent and environmentally friendly method for repairing cracks in concrete. To improve on this ability of microbial materials concrete repair, we applied random mutagenesis and optimization of mineralization conditions to improve the quantity and crystal form of microbially precipitated calcium carbonate. *Sporosarcina pasteurii* ATCC 11859 was used as the starting strain to obtain the mutant with high urease activity by atmospheric and room temperature plasma (ARTP) mutagenesis. Next, we investigated the optimal biomineralization conditions and precipitation crystal form using Plackett-Burman experimental design and response surface methodology (RSM). Biomineralization with .73 mol/l calcium chloride, 45 g/l urea, reaction temperature of 45C, and reaction time of 22 h, significantly increased the amount of precipitated calcium carbonate, which was deposited in the form of calcite crystals. Finally, the repair of concrete using the optimized biomineralization process was evaluated. A comparison of water absorption and adhesion of concrete specimens before and after repairs showed that concrete cracks and surface defects could be efficiently repaired. This study provides a new method to engineer biocementing material for concrete repair.

Summary:

The effect of different factors on the amount of precipitated calcium carbonate and the calcite ratio were shown in Fig. The model predicted that when the concentration of calcium chloride was .73 mol/l, the concentration of urea was 45 g/l, and the temperature was 4C, the amount of precipitated calcium carbonate was 68.8 g/l. The results of XRD and FTIR analysis were consistent, indicating that the precipitate obtained under condition 1 contained both calcite and vaterite crystals, while the precipitate obtained under condition 2 contained only calcite crystals. Repair Effect of Microbially Precipitated Calcium Carbonate on Concrete Specimens The optimized mineralization conditions were used to repair small defects on the surface of the concrete specimens and large-sized cracks to detect the repair effect. Thus, the method developed in this study was able to repair larger cracks in concrete specimens. The water absorption of concrete specimens was tested after repair. 2 A), it did not lead to a further increase of calcium carbonate precipitation 28, 29. During microbially induced calcium carbonate deposition, microbial urease decomposes urea to produced CO₂.

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Culture, *S. pasteurii* ATCC 11859, b, bacterial cells, cell, cells, solid bacteria

Organisms: N/A

Tissues: specimens

Pathways: N/A

Query Result 7/25

Isolation of alkaliphilic calcifying bacteria and their feasibility for enhanced CaCO₃ precipitation in biobased cementitious composites

Shaheen, Nafeesa; Jalil, Amna; Adnan, Fazal; Arsalan Khushnood, Rao

Microbial Biotechnology

Year: 2021 PMID: 33629805 doi: 10.1111/1751-7915.13752 Citations: 0 Score: 13.80

Keywords hits: concrete: 24, calcite: 73, precipitation: 44

Abstract:

Microbially induced calcite precipitation (MICP), secreted through biological metabolic activity, secured an imperative position in remedial measures within the construction industry subsequent to ecological, environmental and economical returns. However, this contemporary recurrent healing system is susceptible to microbial depletion in the highly alkaline cementitious environment. Therefore, researchers are probing for alkali resistant calcifying microbes. In the present study, alkaliphilic microbes were isolated from different soil sources and screened for probable CaCO₃ precipitation. Nonureolytic pathway (oxidation of organic carbon) was adopted for calcite precipitation to eliminate the production of toxic ammonia. For this purpose, calcium lactate Ca(C₃H₅O₃)₂ and calcium acetate Ca(CH₃COO)₂ were used as CaCO₃ precipitation precursors. The quantification protocol for precipitated CaCO₃ was established to select potent microbial species for implementation in the alkaline cementitious systems as more than 5% of isolates were able to precipitate CaCO₃. Results suggested 8% of potent calcifying strains isolated in this study, portrayed higher calcite precipitation at pH 1 when compared to pH 7. Ten superlative morphologically distinct isolates capable of CaCO₃ production were identified by 16SrRNA sequencing. Sequenced microbes were identified as species of *Bacillus*, *Arthrobacter*, *Planococcus*, *Chryseomicrobium* and *Corynebacterium*. Further, microstructure of precipitated CaCO₃ was inspected through scanning electron microscopy (SEM), Xray diffraction (XRD) and thermal gravimetric (TG) analysis. Then, the selected microbes were investigated in the cementitious mortar to rule out any detrimental effects on mechanical properties. These strains showed maximum of 36% increase in compressive strength and 96% increase in flexural strength. *Bacillus*, *Arthrobacter*, *Corynebacterium* and *Planococcus* genera have been reported as CaCO₃ producers but isolated strains have not yet been investigated in conjunction with cementitious mortar. Moreover, species of *Chryseomicrobium* and *Glutamicibacter* were reported first time as calcifying strains.

Summary:

2 Quantification of calcite precipitation by calcifying bacterial strains under different calcium sources, i. e. calcium lactate and calcium acetate at the incubation age of 7 and 14 days. Effect of varying p H on calcite precipitation The selected strains were tested for calcite production at p H 7 and p H 1 with calcium lactate as calcium source. 3 Calcite precipitation of selected calcifying strains at p H 7 and p H 1 with calcium lactate as calcium source. Strains 6 C and 6 E presented similar calcite precipitation at both values of p Hs, while 3 A and 4 F showed higher activity at p H 7. Vaterite and calcite precipitation is bacterial strain and calcium sourcedependent. *Bacillus* species are extensively researched in selfhealing concrete however, isolated strains in the recent studies have not been reported yet. *B. subtilis* was used as a reference control. Effect of varying p H on calcite precipitation To check the effect of p H on calcite production, the selected 1 calcifying bacterial strains were inoculated in CPM media at p H 7 and p H 1 for 14 days.

Mentioned biomedical entities:

Genes: 4A, Lysozyme, TG, tin

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 4B, C, C upriavidus, Cem Concr, E., J., Spore, Tstat, cell, flexural, spore, subtilis

Organisms: Adv Mat Sci, *Bacillus subtilis*, *E. coli*, Ferris,, H. M.

Tissues: lens, specimens, tubes

Pathways: N/A

Query Result 8/25

A review on the potential of filamentous fungi for microbial self-healing of concrete

Van Wylick, Aurlie; Monclaro, Antonielle Vieira; Elsacker, Elise; Vandeloock, Simon; Rahier, Hubert; De Laet, Lars; Cannella, David; Peeters, Eveline

Fungal Biology and Biotechnology

Year: 2021 PMID: 34794517 doi: 10.1186/s40694-021-00122-7 Citations: 0 Score: 13.43

Keywords hits: concrete: 13, calcite: 17, precipitation: 34

Abstract:

Concrete is the most used construction material worldwide due to its abundant availability and inherent ease of manufacturing and application. However, the material bears several drawbacks such as the high susceptibility for crack formation, leading to reinforcement corrosion and structural degradation. Extensive research has therefore been performed on the use of microorganisms for biologically mediated self-healing of concrete by means of CaCO_3 precipitation. Recently, filamentous fungi have been recognized as high-potential microorganisms for this application as their hyphae grow in an interwoven three-dimensional network which serves as nucleation site for CaCO_3 precipitation to heal the crack. This potential is corroborated by the current state of the art on fungi-mediated self-healing concrete, which is not yet extensive but valuable to direct further research. In this review, we aim to broaden the perspectives on the use of fungi for concrete self-healing applications by first summarizing the major progress made in the field of microbial self-healing of concrete and then discussing pioneering work that has been done with fungi. Starting from insights and hypotheses on the types and principles of biomineralization that occur during microbial self-healing, novel potentially promising candidate species are proposed based on their abilities to promote CaCO_3 formation or to survive in extreme conditions that are relevant for concrete. Additionally, an overview will be provided on the challenges, knowledge gaps and future perspectives in the field of fungi-mediated self-healing concrete.

Summary:

Given that certain fungi and bacteria have a native capability of CaCO_3 -producing biomineralization, they are considered to be efficient strengthening 28 or self-healing 29, 3 agents and this has given rise to the development of applications in which microbial cells and nutrients are added to the concrete mix prior to its use, as a prevention measure. Therefore, and because of the requirements for the self-healing function, the characteristics of the selected species or strain for use as a self-healing agent in concrete should meet the following stringent criteria: it should be a non-pathogenic strain, it should be a sporulating strain producing highly resistant spores, it should be capable of vegetative growth in conditions relevant for crack formation, it should be capable of surviving and growing in a harsh alkaline environment and it should promote the precipitation of CaCO_3 at the cell wall surface. Fig. However, existing structures can be targeted as well, in which case cracks could be healed by means of a repair mortar or spray containing fungal spores, as already done with bacteria. Indeed, the large surface area of fungal cells and the presence of negatively charged structures in their cell wall offer multiple metal-binding active sites to bind metal cations 66, 67. Many species are also able to thrive in environments that are highly polluted by metals, such as the fungi used as a biosorbent for heavy metals removal from wastewater 5, 66. Hypotheses underlying a high potential of fungi for self-healing of concrete Based on the initial observations with respect to the use of *T. reesei*, *A. nidulans* and *F. oxysporum* for concrete self-healing 5, 23, 35, and the more extensive knowledge on bacteria-based self-healing concrete 2, 11, 1622, conceptual principles and hypotheses can be drawn for the use of filamentous fungi in these applications. Fungal species with lifestyles that are compatible with the concrete environment could include alkaliphilic, ureolytic and/or oligotrophic species. Table 1 Overview of eligible candidates for fungi-mediated self-healing concrete Fungal species Interesting characteristics Source *Aspergillus nidulans*, *Trichoderma reesei*, *Fusarium oxysporum* H regulatory mutant Grows on concrete layer CaCO_3 precipitation 5, 23, 35 *Aspergillus* sp., *Aureobasidium* sp., *Cephalosporium* sp., *Fusarium* sp., *Monilla* sp. Grows on limestone rock Epi- and endolithic fungi 72 *Cephalotrichum oligotrophicum*, *Chrysosporium guizhouense*, *Clonostachys rosea* Grows in limestone cave Oligotrophic fungi 81 *Mortierella* sp. Grows on moonmilk Oligotrophic fungi 51 *Paecilomyces inflatus*, *Plectosphaerella cucumerina* Grows on calthemite straw stalactite growing from a concrete ceiling Oligotrophic fungi 83 *Chrysosporium* sp., *Paecilomyces lilacinus* Grows on media with a maximum pH of 11 Alkaliphilic fungi 23 *Neurospora crassa*, *Penicillium chrysogenum* CS1 CaCO_3 precipitation in the presence of urea 54, 63 A special focus can be placed on alkaliphiles, which grow well at pH values higher than pH 9, with optimal growth typically observed at a pH value between 1 and 13, whereas slow or no growth is observed at near-neutral pH values 78, 79.

Mentioned biomedical entities:

Genes: CaM, ER

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: *Chrysosporium*, DC, Fungi, HR, *Penicillium*, *Penicillium chrysogenum* CS1 CaCO_3 , SEM, b germinating, b, cell, cell wall surface, cells, fibres,

filamentous cellular, limestone caveOligotrophic fungi81Mortierella, material, spore

Organisms: roots

Tissues: layer, mineral, specimens

Pathways: N/A

Query Result 9/25**Carbonate Precipitation through Microbial Activities in Natural Environment, and Their Potential in Biotechnology: A Review**

Zhu, Tingting; Dittrich, Maria

Frontiers in Bioengineering and Biotechnology

Year: 2016

PMID: 26835451

doi: 10.3389/fbioe.2016.00004

Citations: 0

Score: 12.85

Keywords hits: concrete: 3, calcite: 26, precipitation: 83

Abstract:

Calcium carbonate represents a large portion of carbon reservoir and is used commercially for a variety of applications. Microbial carbonate precipitation, a by-product of microbial activities, plays an important metal coprecipitation and cementation role in natural systems. This natural process occurring in various geological settings can be mimicked and used for a number of biotechnologies, such as metal remediation, carbon sequestration, enhanced oil recovery, and construction restoration. In this study, different metabolic activities leading to calcium carbonate precipitation, their native environment, and potential applications and challenges are reviewed.

Summary:

Microbial carbonate precipitation can occur as a by-product of common microbial metabolic activities, including photosynthesis, ureolysis, denitrification, ammonification, sulfate reduction, and methane oxidation. In freshwater and marine environments, the photosynthesis by cyanobacteria is the most common metabolic process for carbonate precipitation, followed by sulfate reduction. In hot-spring environments, only photosynthesis by cyanobacteria is reported. Caves are characterized by high humidity and relatively low and stable temperature. Both phototrophs and heterotrophs can increase pH through their metabolic activity, and the alteration of the saturation index in mats can represent a combined community effort. According to Ludwig et al., the increase of Ca^{2+} concentration is mainly due to the EPS degradation by SRB, and the driving factor of carbonate precipitation lies in the increased concentration of CO_3^{2-} by photosynthetic bacteria. Therefore, with the injection of a high concentration of dissolved calcium, both inorganic carbon and alkalinity have to be sufficiently produced in situ. Most of the biogrout studies were based on calcium carbonate precipitation induced by hydrolysis of urea. The biomineralization of ureolytic bacteria *S. pasteurii* has significantly increased the soil strength.

Mentioned biomedical entities:

Genes: Arp, MCP, MPa, Sen

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, Cell, Cell walls, Cells, Chlorella, MCP, MPa, Miller, Newman, Nitrosomonas, Oil, Penicillium, SRB cells, biofilm, cell, cells, cereus, cultures, grains, microbes, oil, oil wells, pili, solid substratum, sphaericus strains, wells, wild-type cells

Organisms: *A. variabilis*, *Bacillus cereus*, *Bacillus subtilis*, Bang, human, stone

Tissues: high-permeability zones, leaves, mineral, oil-and-rock, pulp, tubular

Pathways: N/A

Query Result 10/25

A comprehensive screening on the effect of *Bacillus cereus* and *Vibrio natriegens* bacterial consortia in cement mortar
B, Sunantha; Nagarajan, Praveen; Sudhakumar, J.; Thomas, Blessen Skariah; Sankaranarayanan, Pooja; Pitchaikani, Sasikumar;
T.V, Suchithra

Scientific Reports

Year: 2025 PMID: 40595441 doi: 10.1038/s41598-025-94641-7 Citations: 0 Score: 12.83

Keywords hits: concrete: 82, calcite: 12, precipitation: 2

Abstract:

Self-healing mechanisms based on calcite precipitation have proven effective in enhancing concretes strength, durability, and crack repair capabilities. This research explores the self-healing process in concrete via microbial-induced calcium carbonate precipitation (MICP) using *Bacillus cereus* and *Vibrio natriegens*. This study involved various experimental analysis such as biochemical and morphological identification, bacterial growth evaluation, and characterization of mortar specimens using energy-dispersive X-ray spectroscopy (EDS), X-ray diffraction (XRD), testing for ultrasonic pulse velocity (UPV) and compressive strength testing. The findings revealed that the bacterial consortium outperformed the individual bacterial strains, resulting in a 19.85% increase in compressive strength and a 29.6% decrease in water absorption compared to conventional mortar. XRD and EDS analyses confirmed the formation of calcium carbonates within the cracks. These results position the bacterial consortium as an effective and economically viable approach for bio-self-healing concrete through MICP, especially in hot and humid environments.

Summary:

1.159/1-3765217216617.284891991.159/1-376521721661746. Pinto PDe RCristina FBrandao YBSarubbo LABenachour MBiomineralization of calcium carbonate by *Bacillus*, bc toggle=yes*scereus* for self-healing biocement DOAJ221.333/cet27917 Pinto, P. et al. Biomineralization of calcium carbonate by *Bacillus cereus* for self-healing biocement. The findings are expected to advance concrete technology and expand its potential applications in the construction industry. Materials and methods In this study, a series of bacterial and material characterization techniques were done to evaluate the potential of *Vibrio natriegens* and its consortium with *Bacillus cereus* for enhancing bio mineralization in cement mortar. In this study, the bacterial cultures of *Bacillus cereus* and *Vibrio natriegens* showed maximum growth at 16 and 14 concentrations, which were fixed for the preparation of the bacterial solutions. Table 4 Dilution results of *B. cereus* and *V. natriegens*. Dilution Factor111213141516171819 *B. cereus*. Compared with the control mortar, water absorption was decreased by 9.6% for mortar with *Bacillus cereus*, 23% for mortar with *Vibrio natriegens*, and 29.6% for consortia bacteria mortars. The presence of calcium carbonate in the bacterial sample is higher than that of the control sample55.

Mentioned biomedical entities:

Genes: SHC

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Biofilm, *C. E.*, MTP, *b.*, *b.* EDS, *b.* Quadrant, bacterial cells, biofilm, cell, cells, *cereus*, colonies, layer, spore, *subtilis*Organisms: *A. A.*, *Bacillus cereus*, *Bacillus sphaericus*, *C. S.*, *E. I.*, *E. coli*, *Escherichia coli*, *W. M. F.*, *bacillus cereus*, cement, humanTissues: *J.*, inner core, layer, mineral, plastic tissue, specimens, tube

Pathways: N/A

Query Result 11/25

Microbialinduced carbonate precipitation (MICP) technology: a review on the fundamentals and engineering applications

Zhang, Kuan; Tang, Chao-Sheng; Jiang, Ning-Jun; Pan, Xiao-Hua; Liu, Bo; Wang, Yi-Jie; Shi, Bin

Environmental Earth Sciences

Year: 2023

PMID: 37128499

doi: 10.1007/s12665-023-10899-y

Citations: 0

Score: 11.96

Keywords hits: concrete: 56, calcite: 12, precipitation: 24

Abstract:

The microbialinduced carbonate precipitation (MICP), as an emerging biomineralization technology mediated by specific bacteria, has been a popular research focus for scientists and engineers through the previous two decades as an interdisciplinary approach. It provides cutting-edge solutions for various engineering problems emerging in the context of frequent and intense human activities. This paper is aimed at reviewing the fundamentals and engineering applications of the MICP technology through existing studies, covering realistic need in geotechnical engineering, construction materials, hydraulic engineering, geological engineering, and environmental engineering. It adds a new perspective on the feasibility and difficulty for field practice. Analysis and discussion within different parts are generally carried out based on specific considerations in each field. MICP may bring comprehensive improvement of static and dynamic characteristics of geomaterials, thus enhancing their bearing capacity and resisting liquefaction. It helps produce eco-friendly and durable building materials. MICP is a promising and cost-efficient technology in preserving water resources and subsurface fluid leakage. Piping, internal erosion and surface erosion could also be addressed by this technology. MICP has been proved suitable for stabilizing soils and shows promise in dealing with problematic soils like bentonite and expansive soils. It is also envisaged that this technology may be used to mitigate against impacts of geological hazards such as liquefaction associated with earthquakes. Moreover, global environment issues including fugitive dust, contaminated soil and climate change problems are assumed to be palliated or even removed via the positive effects of this technology. Bioaugmentation, biostimulation, and enzymatic approach are three feasible paths for MICP. Decision makers should choose a compatible, efficient and economical way among them and develop an on-site solution based on engineering conditions. To further decrease the cost and energy consumption of the MICP technology, it is reasonable to make full use of industrial by-products or wastes and non-sterilized media. The prospective direction of this technology is to make construction more intelligent without human intervention, such as autogenous healing. To reach this destination, MICP could be coupled with other techniques like encapsulation and ductile fibers. MICP is undoubtedly a mainstream engineering technology for the future, while ecological balance, environmental impact and industrial applicability should still be cautiously treated in its real practice.

Summary:

More specifically, this paper covers the applications of MICP in geotechnical engineering, construction materials, hydraulic engineering, geological engineering, and environmental engineering with a focus on the implementation, efficiency, cost/viability and environmental impacts of the MICP technology in these fields. In this part, several typical applications of the MICP technology in geotechnical engineering are reviewed with a focus on the extent to which this technology can improve the macro-scale mechanical properties of geomaterials. Enhancing soil bearing capacity Like many other thriving engineering technologies, the MICP technology is faced with the challenge of up-scaling implementation when it comes to improving the mechanical properties of soils, such as bearing capacity. They reached an improvement of 37% for the compressive strength of specimens. Applying .6 g of Ca per cm² of sand surface, the permeability of the sand could be reduced from 14 m/s to 1.617 m/s due to formation of the crust on sand surface. Thermal conductivity of MICP treated soils was increased by 25% as compared with the untreated soils.

Mentioned biomedical entities:

Genes: LL, Yao

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AC, Bag MeasurementsPseudomonas, DF, F, F GGNN, Le Pape, MICP cement/concrete mortar, Miscanthus sacchariflorus, Sporosarcina pasteurii cells, U. E., b, b, b) Horizontal, biofilm, cell, cells, clay, crust-like calcite-rich layer, fibers, layer, left, less-load, material, oil, spore

Organisms: Bacillus sphaericus, Bai, fish, human, human urine, stone

Tissues: capillary, cone, grains, high-permeability zones, layer, specimens, transverse cracks

Pathways: N/A

Query Result 12/25

Nanoscale, bChar: bterization of Fungal-Induced CaCO₃ Precipitation:, bImp: bations for Self-Healing Concrete
Tuyishime, J. R. Marius; Hammer, Edith C.; Pla-Ferriol, Mart; Thnell, Karina; Alwmark, Carl; van Velzen, Sophie; Floudas, Dimitrios;
Platakyte, Rasa; Obst, Martin; Zou, Hanbang

ACS Applied Materials Interfaces

Year: 2025 PMID: 40542343 doi: 10.1021/acsami.5c07137 Citations: 0 Score: 11.85

Keywords hits: concrete: 24, calcite: 23, precipitation: 18

Abstract:

Cracks in concrete, bUniv: bsity, 223 62, bsity. From these samples, three different urease-positive ascomycetes, bcomp: bmise structural integrity by exposing, bnds.

Summary:

The Ca spectra,, bA and A illustrate, b). The X-ray transmission photons were detected by, ba P4: bscintillator attached to a photomultiplier tube detector. To obtain chemical information, ultrathin sections were raster-scanned, bacro: b a nanofocused X-ray beam in the x and y directions at discrete, b all five spots suggest possible variations, binci: bnt energies tuned around the X-ray absorption edges of Ca, be V): bd C K -edge, with energy steps of between, b.1: b211.2 e V around the absorption peaks to capture detailed, binfo: bation and relatively larger steps elsewhere. , b thin-section areas, particularly in spot b (Figure , bs align with the Ca L-edge data, further confirming the predominance, bThe: bnction performs spectrum-by-spectrum fitting to determine the, bvestigated thin sections were coated with a 7 nm thick layer, besence of a sharp peak at 29.3 e V, and a smaller, more, bminance of the calcite polymorph in fungal-induced Ca CO₃ biominerals validates the feasibility of applying fungi in, bopti: bm combination of reference compounds that accurately represent, bze fungal growth conditions and application methods for field-scale, bpure: hydroxyapatite, Ca²⁺ adsorbed on extracellular polymeric, bsubs: bnces, and Ca Cl₂. : Nano2231515415171.139/D2 EN1143 CBindshedler S. Cailleau G. Verrecchia E. Role of Fungi in the Biomineralization, bcite Minerals21662411.339/min6241 Christensen W. In fungal species, b3, C: bEXAFS spectra from two analyzed spots showed additional peaks, barou: b 288.2289.5 e V.

Mentioned biomedical entities:

Genes: L2, L3, P4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, C K-Edge, CaCl₂4H₂O, Fungi, Paracolon Cultures, Proteus, Salmonella, Shigella, b, b 288.2289.5, b 8, b AgesACS, b EPS-rich, b L-edge, b L3,2-edge, b OD, b STXM, b column, b ref, b species, b, cell, cellular

Organisms: bof F, bof b

Tissues: Fine, Soft X-ray Spectromicroscopy, mineral, tube

Pathways: N/A

Query Result 13/25

Mineralogy, morphology, and reaction kinetics of ureolytic bio-cementation in the presence of seawater ions and varying soil materials

Burdalski, Robert J.; Ribeiro, Bruna G. O.; Gomez, Michael G.; Gorman-Lewis, Drew

Scientific Reports

Year: 2022 PMID: 36224231 doi: 10.1038/s41598-022-21268-3 Citations: 0 Score: 11.56

Keywords hits: concrete: 1, calcite: 31, precipitation: 46

Abstract:

Microbially-induced calcium carbonate precipitation (MICP) is a bio-cementation process that can improve the engineering properties of granular soils through the precipitation of calcium carbonate (CaCO_3) minerals on soil particle surfaces and contacts. The technology has advanced rapidly as an environmentally conscious soil improvement method, however, our understanding of the effect of changes in field-representative environmental conditions on the physical and chemical properties of resulting precipitates has remained limited. An improved understanding of the effect of subsurface geochemical and soil conditions on process reaction kinetics and the morphology and mineralogy of bio-cementation may be critical towards enabling successful field-scale deployment of the technology and improving our understanding of the long-term chemical permanence of bio-cemented soils in different environments. In this study, thirty-five batch experiments were performed to specifically investigate the influence of seawater ions and varying soil materials on the mineralogy, morphology, and reaction kinetics of ureolytic bio-cementation. During experiments, differences in reaction kinetics were quantified to identify conditions inhibiting CaCO_3 precipitation and ureolysis. Following experiments, scanning electron microscopy, x-ray diffraction, and chemical composition analyses were employed to quantify differences in mineralogical compositions and material morphology. Ions present in seawater and variations in soil materials were shown to significantly influence ureolytic activity and precipitate mineralogy and morphology, however, calcite remained the predominant CaCO_3 polymorph in all experiments with relative percentages exceeding 8% by mass in all precipitates.

Summary:

All treatment solutions were prepared more concentrated in order to achieve targeted concentrations after dilution with augmented soils, which contained 1 mL solution volumes from cell inoculants. Aqueous solution sampling Small-volume aqueous solution samples were collected from all experiments at various times during reactions to monitor changes in solution urea and Ca^{2+} concentrations reflective of urea hydrolysis activity and CaCO_3 precipitation. When comparing ureolytic activity in the 54 mM Mg^{2+} with 1 mM Ca^{2+} experiment to the 1% ASW specimen, however, ureolytic rates were nearly identical, suggesting that the inhibition of ureolysis observed in 1% ASW may primarily result from the presence of 54 mM Mg^{2+} . Although ureolytic inhibition observed in the 1% ASW experiments were primarily attributed to the 54 mM Mg^{2+} concentrations, the ionic strength of the 1% ASW solutions were near 1458 mM when 25 mM Ca^{2+} was supplied. All experiments included the addition of 25 mM urea, 25 mM Ca^{2+} , *S. pasteurii* cells at 8.17 cells/mL and varying added Na^+ concentrations. Soil varied experiments In order to investigate the effect of changes in soil materials, a series of experiments were performed using eleven different soils. When geochemical conditions similar to those considered in this study are present, CaCO_3 precipitation during MICP may be assumed to be an equilibrium reaction and eliminate the need for other computationally intensive kinetic expressions for CaCO_3 precipitation to be incorporated in geochemical models. Quantitative comparisons of activity-based cell density estimates between all experiments and their respective controls suggested that the most significant inhibition of ureolysis occurred in experiments containing 1% ASW, 18 mM Mg^{2+} , and Fraser River Sand with ureolytic activities exceeding rates observed in the control experiments in only the montmorillonite and mica containing specimens.

Mentioned biomedical entities:

Genes: E13, E16, E21, E25, E4, MP, NSF

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Aggregates, CaCO_3 384, Cell, Concentrated *S. pasteurii* cell pellet, *S. pasteurii* cell, *S. pasteurii* cell suspensions, *S. pasteurii* cells, activities98.Figure 2Relative CaCO_3 , b, b) Ca^{2+} , b) Mg^{2+} , b) S, b) Sr^{2+} , b, c) montmorillonite, b, d) Ca^{2+} , cell, cell suspensions, cells, cellular, concentrate cells, electrolyte, freeze-dried cell pellets, material, *pasteurii* cell, *pasteurii* cells, specimens, ureolytic cell

Organisms: mineral

Tissues: mineral, specimens, tube

Pathways: N/A

Query Result 14/25

Evaluation and optimal width ratio selection of microbial mineralization technique in the repair of lining cracks in Xinjiang desert open channel

Wang, Jianxin; Zhu, Jianing; Li, Yusheng; Zhang, Shihao; Feng, Chengming

Scientific Reports

Year: 2025 PMID: 40374753 doi: 10.1038/s41598-025-01582-2 Citations: 0 Score: 11.50

Keywords hits: concrete: 65, calcite: 6, precipitation: 2

Abstract:

The bank slope structure of the open channel in the Xinjiang desert is affected by seasonal climate changes and water erosion, leading to lining cracking. This study identified the optimal culture conditions and mineralization factors for *Sporosarcina pasteurii* through strain cultivation and precipitation assays. Subsequently, .15mm wide penetrating concrete cracks were prefabricated, and microbially induced calcium carbonate precipitation (MICP) repair tests were conducted over 16 cycles. These experiments included macroscopic performance evaluations, such as area repair rate, penetration resistance, and capillary water absorption tests, alongside microscopic analyses using X-ray diffraction and scanning electron microscopy. The results indicate that MICP technology effectively repairs narrow cracks, preserving crack surface integrity, significantly reducing permeability and water absorption, and enhancing the durability of the concrete. However, for cracks exceeding 1mm in width, the repair efficiency declines progressively. Based on the experimental data, it is concluded that a crack width of 1mm is the optimal threshold for effective MICP-based repair within 16 cycles, ensuring both structural integrity and optimal waterproofing. These results offer valuable insights into the potential application of MICP technology for the remediation of lining cracks in the bank slopes of water conveyance channels in Xinjiang Desert.

Summary:

The designed strength grade of the concrete was C2 based on the strength of the lining of the bank slope of the desert open channel^{27,28}. In the study of microbial mineralization restoration, according to the thickness of the bank slope lining is 6 cm, a concrete cylinder with dimensions of 1 mm 6 mm was utilized, featuring crack lengths and depths of 5 mm and 6 mm, respectively, with crack widths ranging from .1 to 5. mm, 12 test pieces in total. 2 Preparation process of microbial concrete. Rehabilitation assessment program The cracks were treated using MICP over a duration of 16 d. During this period, the area repair rate, permeability coefficient recovery rate and capillary water absorption recovery rate of concrete specimens with different crack widths and repair time were observed. To establish Width for concrete crack repair only through MICP, a series of 16 consecutive cycles of MICP repair were conducted on concrete cracks with widths varying from .1 to 5. mm. The area repair rate test was conducted initially, with the repair of certain specimens exhibiting width cracks in. Fig. 6 Crack area repair rate at different repair cycles/crack widths. Analysis in Fig. This fitting formula was developed to serve as a reference for the area repair of cracks in desert channel linings Penetration test The variation in water penetration before and after the repair of cracks in concrete specimens indicates that the recovery rate of permeability coefficient can serve as a quantitative measure of the effectiveness of crack repair, with a total of 39 specimens tested.

Mentioned biomedical entities:

Genes: BLC

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: SiO₂, b, b indicates, b presents SEM, b, layer, material

Organisms: *Bacillus subtilis*

Tissues: capillary, capillary pores, capillary water, infiltrate, inner walls, layer, smooth, specimens

Pathways: N/A

Query Result 15/25

Hard shell, soft blue-green core: Ecology, processes, and modern applications of calcification in terrestrial cyanobacteria
Jung, Patrick; Briegel-Williams, Laura; Dultz, Stefan; Neff, Carina; Heibrock, Gunnar; Monger, Curtis; Pietrasiak, Nicole; Keller, Lena;
Hale, Julia; Friedek, Jan; Schmidt, Timo; Guggenberger, Georg; Lakatos, Michael

iScience

Year: 2024

PMID: 39628580

doi: 10.1016/j.isci.2024.111280

Citations: 0

Score: 11.42

Keywords hits: concrete: 8, calcite: 35, precipitation: 33

Abstract:

Cyanobacteria are the oldest photoautotrophic lineage that release oxygen during photosynthesis, an ability that possibly evolved as far as 3.5 billion years ago and changed the Earth's environment both in water and on land. Linked to the mechanism of carbon accumulation by cyanobacteria during photosynthesis are their calcifying properties, a process of biologically mediated mineralization of CO₂ by precipitation with calcium to CaCO₃. In recent decades, scientific research has mainly focused on calcifying cyanobacteria from aquatic habitats, while their terrestrial relatives have been neglected. This review not only presents the ecology of terrestrial calcifying cyanobacteria in caves and biocrusts but also discusses recent biotechnological applications, such as the production of living building materials through microbial-induced carbonate precipitation for structural engineering, which has the potential to open a new and efficient pathway for mitigating climate change, e.g., as carbon capture and storage technology.

Summary:

Scytonema sp. And Scytonema sp. However, induction of MICP in desert soils as a mechanism to increase soil stability requires further investigation, standardization, and optimization.⁴³ Nevertheless, applications preventing desertification based on biocrust cyanobacteria are on the rise, but they are usually linked to the filamentous structure of the cyanobacteria and the gluing properties of their EPS rather than on calcification processes.¹⁵⁸ The use of bio-mediated calcium carbonates, for example, has been proposed as a useful bio-protection material for weathered rocks.¹⁵⁹ Most of these scenarios are inspired by natural biocrusts, which, for example, also stabilize the Great Wall of China against erosion.¹⁶ A common methodology for using cyanobacteria in erosion prevention is isolating a natural community from, e. g., biocrusts, cyanobacterial upscaling in the laboratory, and the artificial application of the biomass to soil surfaces prone to erosion. Heterotrophic bacteria utilize organic compounds such as urea, NO₃, amino acids, or methane for energy and cell material.^{167,168} An increase in the pH of the surrounding pore water is caused by the metabolic activity of the bacteria, resulting in supersaturated conditions and inducing the precipitation of carbonates in the presence of an inorganic calcium source. The production of LBMs, for example, is currently restricted to the utilization of a single, unicellular marine model strain however, it is surprisingly not known if it uses calcification in its natural habitat.^{37,38,18} This is in addition to certain drawbacks because this strain needs to be grown in a salt-rich medium, for example, which is problematic for most upscaled bioreactor-based approaches due to corrosion.¹⁹³ In addition, the application of filamentous or even branched cyanobacteria for the manufacturing of LBMs might also lead to a more complex matrix and thus to a higher degree of stability.

Mentioned biomedical entities:

Genes: B, BicA

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: CCM, E., F) Filaments, Florida,²⁴ cell, cells, cells themselves.⁽²⁾ Physical, cellular, concrete-like, cyanobacterial cell, cytosolic, fibers, hollow chambers Cambrian Izabella, layer, material, photosynthesis,⁸⁸ photosynthetic cell, podovirus-mediated cell, unicellular, whitish tubular CaCO₃ sheaths

Organisms: N/A

Tissues: Aerial, bark, bone material, calcified sheaths, calcified sheaths), capillary, connective pore, connective pores, grains, layer, mineral, needle fiber calcite, seed, sheath material, stiff sheaths

Pathways: N/A

Query Result 16/25

Assessment of ecofriendly carbon capture using *Bacillus subtilis* induced calcium carbonate precipitation with focus on applications mechanisms and cost efficiency

Danial, Amal W.; Hasan, Raghad M. M.; Mahmoud, Ghada Abd-Elmonsef; Abdel-Basset, Refat

Scientific Reports

Year: 2025

PMID: 40594560

doi: 10.1038/s41598-025-06688-1

Citations: 0

Score: 11.17

Keywords hits: concrete: 12, calcite: 13, precipitation: 4

Abstract:

This work focuses on exploiting the naturally occurring microbial calcium carbonate precipitation catalyzed by microbial consortia within lakes and oceans biogeochemistry for carbon dioxide removal from atmosphere. In this work, *Bacillus subtilis* OQ119616 was used for carbon dioxide sequestration in equi-molar concentrations into *Bacillus*-induced calcium carbonate precipitation (BICCP). As this process requires alkaline media, urea degradation by urease and nitrogen fixation were traced. BICCP has been formed from calcium salts in the following order: chloridenitrateacetatecitrate. However, conversion efficiency percentage (CE%) of calcium salts to CaCO_3 exhibited a different attitude of citrateacetatechloridenitrate. Calcium citrate is excluded from consideration. Acetate, however, is the most efficient salt; it significantly exhibited the highest CE%, with the least cost and highest economic feasibility. The wide range in quantities, efficiency and feasibility indicates the importance of the salt anion in BICCP. In addition, BICCP exhibited applicability in healing concrete cracks, improving field capacity of sand soil and the subsequently improved seed germination of *Vicia faba*. BICCP was also accompanied by adsorption of heavy metals as partial purging of waste/sewage water for hygiene/reuse. *Bacillus subtilis* exhibited the ability to perform MICP, utilizing various calcium salts in the following order: chlorideacetatenitratecitrate. However, acetate is the most efficient salt of calcium to be converted to calcium carbonate precipitate by *B. subtilis*, as it exhibited the highest conversion efficiency percentage (g/g%), with the least cost and highest economic feasibility. Carbon dioxide removal (CDR) occurs at simultaneous equity to CaCO_3 precipitation at mole/mole ratios. Economic feasibility (US/m³) showed that BICCP may be applicable in CDR for cleansing carbon dioxide inside closed systems and for environmental safety. The bacterially induced CaCO_3 proved successful applicability in improving the field capacity of sand soil and growth of *V. faba*, healing concrete cracks and sorption of heavy metals for depolluting sewage/wastewater for hygiene reuse. BICCP could repair concrete cracks of 12mm wide in 7days by 21×10^6 cells/mL. Adsorption of heavy metals (Pd, Zn, Cd and Cu) for partial removal of contaminants in/from waste/sewage water for hygiene reuse.

Summary:

The dependence of BICCP magnitude as well as its morphs on Ca^{2+} salt type was studied at different concentrations of calcium citrate, acetate, nitrate, or chloride. 4 Calcium carbonate precipitation by *Bacillus subtilis* after growth for 3 days in minimal medium supplemented with four levels of four calcium salts development of calcium carbonate precipitation over time of 3 days at successively increasing concentrations of CaCl_2 Scanning electron micrographs showing shapes and morphogenesis of calcium carbonate precipitations produced by *Bacillus subtilis* at the provided calcium salts 4c Ci, 4c Ac, 4c N and 4c Cl for citrate, acetate, nitrate, and chloride, respectively transmission electron micrographs showing nanoparticles of calcium carbonate precipitations produced by *Bacillus subtilis* at the provided calcium chloride and viability of bacterial cells in calcium carbonate precipitates and. As ammonia results mainly from the activity of the urease enzyme, (Fig. Values of 2 in Table 5 showed that using calcium acetate, calcium citrate and calcium nitrate exerted the large magnitude of effect on the changes of growth parameter in *B. subtilis*, with increasing salt concentrations except in the case of calcium chloride give a small effect with concentration of salt and gave a large magnitude with time and interaction of time and salt concentrations. In this work, scanning electron microscopy displayed four different morphologies of BICCP, which are needle-like produced from calcium nitrate, spherical from calcium citrate, irregular from calcium acetate and rhombohedral for calcium chloride These morphs are valerite according to Chakrabarty and Mahapatra²⁶ and Lee et al.²⁷, aragonite, and calcite according to Barhoum et al.²⁸. BICCP was shown to contain micro-and nanoparticles resulting in compatibility with several applications. The morphologies of calcium carbonate have been repeatedly reported to depend on the type of calcium salt provided, the growth conditions of e. g., pH or temperature as well as on type of the microorganism performing the MICP process, the latter acts as the template or nucleation site for calcium carbonate precipitation.

Mentioned biomedical entities:

Genes: CDR, EF2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: *Bacillus* cells, Culture, Romania).Cells, b, bacterial cells, cell, cells, cultures, needle-like, specimens, *subtilis*, *subtilis* cultures

Organisms: A. W. D., *Bacillus sphaericus*, *Bacillus subtilis*, Human, Lai, humans

Tissues: Mineral, Plumule, pearl, seeds, specimens

Pathways: N/A

Query Result 17/25

A sample cell for the study of enzyme-induced carbonate precipitation at the grain-scale and its implications for biocementation

Zehner, Jennifer; Ryne, Anja; Sikorski, Pawel

Scientific Reports

Year: 2021 PMID: 34211000 doi: 10.1038/s41598-021-92235-7 Citations: 0 Score: 10.98

Keywords hits: concrete: 5, calcite: 68, precipitation: 32

Abstract:

Biocementation is commonly based on microbial-induced carbonate precipitation (MICP) or enzyme-induced carbonate precipitation (EICP), where biomineralization of documentclass12ptminimal, b: bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocumenttext CaCO3enddocumentCaCO3 in a granular medium is used to produce a sustainable, consolidated porous material. The successful implementation of biocementation in large-scale applications requires detailed knowledge about the micro-scale processes of documentclass12ptminimal, bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocumenttext CaCO3enddocumentCaCO3 precipitation and grain consolidation. For this purpose, we present a microscopy sample cell that enables real time and in situ observations of the precipitation of documentclass12ptminimal, bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocumenttext CaCO3enddocumentCaCO3 in the presence of sand grains and calcite seeds. In this study, the sample cell is used in combination with confocal laser scanning microscopy (CLSM) which allows the monitoring in situ of local pH during the reaction. The sample cell can be disassembled at the end of the experiment, so that the precipitated crystals can be characterized with Raman microspectroscopy and scanning electron microscopy (SEM) without disturbing the sample. The combination of the real time and in situ monitoring of the precipitation process with the possibility to characterize the precipitated crystals without further sample processing, offers a powerful tool for knowledge-based improvements of biocementation.

Summary:

The sample volume for one cell is documentclass12ptminimal, bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocument3, upmu text Lenddocument3L and each channel is 17 mm long, 3.8 mm wide, and .4 mm high. The sample cell was used to investigate two different crystallization conditions: EICP in presence of the calcite seeds, and EICP in the presence of sand grains. For both enzyme concentrations only growth of the calcite seeds was detected, and no nucleation of new calcite crystals on the polymer slide was observed. The spatial distribution of pH was investigated further by comparing the average local pH in an area close to a calcite seed (Fig. In the first part of the experiments in the presence of sand, a single injection of the crystallization solution in the sample cell was investigated by monitoring the precipitation process and local pH changes. The local pH differences caused by documentclass12ptminimal, bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocumenttext Ca CO3enddocumentCa CO3 crystals growing on sand surfaces that were not perpendicular to the line of view could not be detected with this method. After a reaction time of 2 h the reaction was stopped by injecting ethanol, the sample cell was disassembled and the polymer slide was washed with DI water and dried for Raman microspectroscopy and SEM analysis. The cross-sectional area of documentclass12ptminimal, bsepackageamsmath, bsepackagewasysym, bsepackageamsfonts, bsepackageamssymb, bsepackageamsbsy, bsepackageamsthfs, bsepackageupgreek, betlengthoddsidemargin-69pt, begindocumenttext Ca CO3enddocumentCa CO3 crystals and aggregates on the surface of the sand grains was measured from the SEM micrographs and is shown as a function of the number of injections. For further injections of the crystallization solution not only the sand grains, but also the calcite crystals need to be considered for the precipitation process.

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, G2, Ibidi sticky-Slide VI, ResultsSample cell, aggregates, begindocument15, upmu text, b,c, b,e, cell, cells, grains, granular medium3, layer, left, line, material, self-adhesive Ibidi sticky-Slide VI, solid, solid calcium carbonate

Organisms: seeds

Tissues: grains, seed, seeds

Pathways: N/A

Query Result 18/25

Finite element simulation of bacterial self-healing in concrete using microstructural transport and precipitation modeling

Vedrtnam, Ajitanshu; Kalauni, Kishor; Palou, M. T.

Scientific Reports

Year: 2025

PMID: 40328859

doi: 10.1038/s41598-025-99844-6

Citations: 0

Score: 10.83

Keywords hits: concrete: 43, calcite: 2, precipitation: 57

Abstract:

Bacteria-based self-healing concrete has emerged as a promising solution for enhancing structural durability by autonomously repairing cracks. However, the underlying transport mechanisms of healing agents and the efficiency of mineral precipitation remain inadequately modelled. This study presents a finite element modelling (FEM) approach to simulate the diffusion and reaction kinetics of self-healing bacterial agents in concrete microstructures. X-ray micro-computed tomography (Micro-CT) finite element meshes were utilized to accurately represent crack and pore geometries, while the diffusion-reaction equation governing calcium carbonate (CaCO_3) precipitation was numerically solved using FEniCS. Key input parameters, including diffusion coefficients, precipitation rates, and healing efficiencies, were extracted from literature to ensure model validation. Simulations reveal that healing agent concentration follows a nonlinear diffusion pattern, with efficiency influenced by crack geometry and bacterial metabolic activity. Heatmaps and contour plots highlight healing agent dispersion, while time-dependent analysis indicates a 65.5% crack closure efficiency under optimal bacterial conditions. The proposed model effectively replicates experimental trends, demonstrating its applicability for predicting healing performance in realistic structural conditions. This study provides a computational framework that can be extended to optimize bacteria encapsulation strategies, healing kinetics, and long-term durability assessments in self-healing concrete.

Summary:

Healing efficiency of bacterial-based self-healing systems has been reported, with crack closure of up to 488% and sealing of cracks as wide as .97 mm, demonstrating the practical viability of MICCP34. The steep closure gradient near the crack opening suggests that healing efficiency is highest in smaller crack widths, where transport and precipitation rates are most effective. , bsecComparison with experimental results from literature and limitations of model The FEM approach used in the manuscript effectively simulates healing agent transport, reaction kinetics, and Ca CO_3 precipitation in bacterial-based self-healing concrete. The 65.5% crack closure efficiency predicted by the model closely aligns with experimental observations, such as those reported by Javeed et al. 5, where bacterial-based healing improved compressive strength by up to 42.8%. A key limitation of the FEM study is its assumption of uniform bacterial activation and ideal diffusion conditions, whereas Bagga et al. 9 found that real-world bacterial performance is highly dependent on environmental factors such as p H, temperature, and bacterial survivability.

Mentioned biomedical entities:

Genes: SHC, fem

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Solid lines, b, b, cellular, high-pH, lines, material

Organisms: N/A

Tissues: capillary, material, mineral, slice

Pathways: N/A

Query Result 19/25

Microbially-Induced-Calcite-Precipitation (MICP): A biotechnological approach to enhance the durability of concrete using *Bacillus pasteurii* and *Bacillus sphaericus*

Nasser, Amal A.; Sorour, Noha M.; Saafan, Mohamed A.; Abbas, Rateb N.

Heliyon

Year: 2022 PMID: 35855981 doi: 10.1016/j.heliyon.2022.e09879 Citations: 0 Score: 10.60

Keywords hits: concrete: 24, calcite: 17, precipitation: 18

Abstract:

Developing bio-based self-healing concrete aims to minimize durability problems related to cracking. In this study, MICP was used as a smart and eco-friendly approach to produce bio-based durable materials. *Bacillus pasteurii* (BP) and *Bacillus sphaericus* (BS) were added into mortar mixtures with .25% and .5% cement weight. All treated samples exhibited a significant decline in water uptake, capillary permeability, and volume of permeable voids, as compared to control with no bacteria. All treated samples showed significant increase in compressive strength by 285%, after 28 days of curing. At the age of 12 days, the flexural strength of all treated samples was significantly increased by 19.2965.94%. SEM imaging and EDAX confirmed that treated samples were denser with less voids due to MICP. DTA verified that the calcite amount and the crystallinity degree were improved in treated samples. Load deflection of bacterial Reinforced-Laminates had less deformation than control. Reloaded bacterial Reinforced-Laminates exhibited excellent restoration of physico-mechanical properties and performance, after 28, 9, and 12 days, confirming the healing process. Microbial self-healing is an innovative approach for continuous repair of micro-cracks in concrete, improving its durability, thus can reduce the maintenance costs.

Summary:

Both species were adapted with same rate and followed similar growth-pattern.25 and .5% of cement weight, after 3, 7, 14, 28, 6, 9, and 12 days.2 Setting time, water absorption rate, capillary permeability, and permeable voids Initial setting time was reduced for all treated samples as compared with control. The overtime reduction in water absorption was also reported 2, 1 as a result of bacterial addition into mortar. Results revealed that, at all tested ages, the permeable voids volume for all treated samples was reduced as compared with control. Authors described nano-deposition of Ca CO₃ through MICP, which probably blocks the micro-cracks and tiny pores in cementitious materials, thus improves its mechanical properties 37, 38, 39, 4. *sphaericus*, and *B. pasteurii* at .25% and .5% concentration of cement weight. Overall, the improvement in crystallinity degree with its stability enhanced the physico-mechanical properties of the produced bio Concrete. *pasteurii*, BS25 and BS5 = *B. sphaericus* at .25% and .5% of cement weight.5 Durability and restoration of mechanical strength The main objective for developing bacterial self-healing is to achieve good healing performance when cracks occur 2. Likewise, reloading have induced new micro-cracks in columns as reported by Al-Tabbaa et al. 46 without reopening formerly healed ones, probably due to full recovery of mechanical strength after self-healing activity. *pasteurii* and *B. sphaericus* at .25% and .5% of cement weight, after 28, 9, and 12 days of curing, compressive strength measurement.6 Durability and restoration of Reinforced-Laminates under load deflection After 28, 9, 12 days, the deflection of bacterial samples was less than control.

Mentioned biomedical entities:

Genes: B

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C, countBacterial cell, lamina, material, megaterium strains, needle-like, subtilis

Organisms: *Bacillus sphaericus*, cement, fish

Tissues: Capillary, capillary, epoxy-resin, lamina, samples

Pathways: N/A

Query Result 20/25

Efficient option of industrial wastewater resources in cement mortar application with river-sand by microbial induced calcium carbonate precipitation

Huang, Yi-Hsun; Chen, How-Ji; Maity, Jyoti Prakash; Chen, Chien-Cheng; Sun, An- Cheng; Chen, Chien-Yen

Scientific Reports

Year: 2020 PMID: 32317706 doi: 10.1038/s41598-020-62666-9 Citations: 0 Score: 10.55

Keywords hits: concrete: 13, calcite: 26, precipitation: 27

Abstract:

The industrial wastewater disposal has been growing attention for environmental protection and resource substitution, current decades. Similarly, the durability enhancement of concrete has increased attention by microbial induced CaCO_3 precipitation (MICP) process (biocalcification). However, ecofriendly utilization of industrial wastewater in concrete formation is unstudied so far. The present study was carried out to evaluate the effect of industrial wastewater on the formation of cement mortar, compressive strength and water absorption. The biocement mortar strength (y) increased ($y = .52952 + 1.619x + 251.5$; $R^2 = .9825$) with increasing percentage of organic wastewater (x) (BMBM1) by MICP, where highest strength (28.75kgf/cm²) was observed on BM1 (1% wastewater), compared to control (252.5kgf/cm²). The water absorption (y) of biocement mortar decreases ($y = .2512.13 + 15.965x$; $R^2 = .9594$) with increment of wastewater (x) (%) (BMBM1), where a minimum-water-absorption (14.42%) observed on BM1, compared to control (15.89%). SEM micrograph and XRD shows the formation of most-distinctive CaCO_3 crystallization (aragonite/calcite) (acicular, brick shape, massive and stacked structure) inside biocement mortar (BM1), which fills the pores within cement mortar to form a denser structure, by microbial organic wastewater. Thus, present findings implied a cost-effective of MICP technology to improve the concrete properties along with the mitigation of industrial wastewater pollution, which goes some way towards solving the problem of industrial wastewater pollution.

Summary:

However, in addition of organic wastewater totally that can effectively improve the strength of cement mortar compare to control. The water absorption of biocement mortar was noticed as 15.89% at 28 days without organic wastewater, whereas 15.51%, 15.58%, 15.6% and 14.87% of water absorption in biocement mortar were observed in the treated group of 2%, 4%, 6% and 8% organic waste water, respectively at 28days. The biocement mortar treated with 1% wastewater was observed a water absorption of 14.42% after 28 days. 7). Conclusion The industrial wastewater was applied to enhance the durability of biocement mortar such as compressive strength, water absorption by microbial-induced calcium carbonate precipitation. The strength of biocement mortar increased and water absorption of biocement mortar decreases with the increasing percentage of organic wastewater by MICP process.

Mentioned biomedical entities:

Genes: BM6, Bar

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: C 2, Line, b, screening44.752.7597.2582.3621.9478.6161.1836.1663.843.651.5148.495.372.6927.311.159.919.92.751..Figure 8Grains

Organisms: Bacillus sphaericus, HI 9828, cement, human

Tissues: Portland cement, biocement mortar cube, biocement mortar cube particles, cement mortar samples, grains, mineral

Pathways: N/A

Query Result 21/25**Influencing factors on ureolytic microbiologically induced calcium carbonate precipitation for biocementation**

Erdmann, N.; Strieth, D.

World Journal of Microbiology Biotechnology

Year: 2022

PMID: 36576609

doi: 10.1007/s11274-022-03499-8

Citations: 11

Score: 10.53

Keywords hits: concrete: 2, calcite: 1, precipitation: 29

Abstract:

Microbiologically induced calcium carbonate precipitation (MICP) is a technique that has received a lot of attention in the field of geotechnology in the last decade. It has the potential to provide a sustainable and ecological alternative to conventional consolidation of minerals, for example by the use of cement. From a variety of microbiological metabolic pathways that can induce calcium carbonate (CaCO₃) precipitation, ureolysis has been established as the most commonly used method. To better understand the mechanisms of MICP and to develop new processes and optimize existing ones based on this understanding, ureolytic MICP is the subject of intensive research. The interplay of biological and civil engineering aspects shows how interdisciplinary research needs to be to advance the potential of this technology. This paper describes and critically discusses, based on current literature, the key influencing factors involved in the cementation of sand by ureolytic MICP. Due to the complexity of MICP, these factors often influence each other, making it essential for researchers from all disciplines to be aware of these factors and its interactions. Furthermore, this paper discusses the opportunities and challenges for future research in this area to provide impetus for studies that can further advance the understanding of MICP.

Summary:

PCC 7316 Synechococcus pevalekii BDHKU 3511 Tolerance of most strains to alkaline environment No toxic by-products Cheap media for cultivation CO₂ limitation in sand Light dependent Slow growth rates during phototrophic cultivation Dhami et al., Zhu et al. Ureolysis Sporosarcina pasteurii Bacillus sphaericus Bacillus cereus Bacillus megaterium Easy to control Cheap media components High calcium carbonate precipitation rates High tolerance of strains to alkaline environment Toxic ammonia as by-product Production of urea coupled to CO₂ emissions Hammes and Verstraete, Muynck et al., Stocks-Fischer et al. Denitrification Castellianella denitrificans Pseudomonas denitrificans By-products calcium formate and calcium nitrate are commercial additives for cement Nitrite ions can inhibit corrosion Usage of carriers for alkaline environments necessary Eran et al. Beyond the urea metabolism of the microorganisms, the calcium metabolism also has an influence on the MICP. Therefore, it will not only be necessary to find suitable strains depending on ureolytic activity, pH tolerance or the ability to achieve high biomass concentrations during cultivation. For example, Sotoudehfar et al. found an optimum compressive strength of sand at a ratio of urea to calcium chloride of 3 M to 1.5 M. Muynck et al. also investigated the influence of different concentrations of urea and calcium on MICP. This is especially true since compressive strength of biocemented soil correlates with the amount of Ca CO₃ after treatment and the amount of precipitated Ca CO₃ during MICP is directly dependent on the initial concentration of urea and calcium ions and the number of treatment cycles of cell suspension and cementation solution. Using this method, Cheng et al. were able to consolidate silica sand to a compressive strength of 2.5 MPa using Sporosarcina pasteurii and achieved a nearly homogeneous distribution in a 36 mm column.

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 222)Pseudomonas, C, Cell, MF-2, MPa, Mitchell, Smooth, TU, b, bacterial cells, cell, cells, specimens

Organisms: Bacillus megaterium, Escherichia coli, Lai, human, humans, pigs, sand bed

Tissues: grain, specimens, zones

Pathways: N/A

Query Result 22/25

Agricultural by-products and oyster shell as alternative nutrient sources for microbial sealing of early age cracks in mortar

Hong, Minyoung; Jang, Indong; Son, Yongjun; Yi, Chongku; Park, Woojun

AMB Express

Year: 2021 PMID: 33409575 doi: 10.1186/s13568-020-01166-5 Citations: 0 Score: 10.39

Keywords hits: concrete: 47, calcite: 11, precipitation: 14

Abstract:

Bio-concrete using bacterially produced calcium carbonate can repair microcracks but is still relatively expensive due to the addition of bacteria, nutrients, and calcium sources. Agricultural by-products and oyster shells were used to produce economical bio-concrete. Sesame meal was the optimal agricultural by-product for low-cost spore production of the alkaliphilic *Bacillus miscanthi* strain AK13. Transcriptomic dataset was utilized to compare the gene expressions of AK13 strain under neutral and alkaline conditions, which suggested that NaCl and riboflavin could be chosen as growth-promoting factors at alkaline pH. The optimal levels of sesame meal, NaCl, and riboflavin were induced with the central composite design to create an economical medium, in which AK13 strain formed more spores with less price than in commercial sporulation medium. Calcium nitrate obtained from nitric acid treatment of oyster shell powder increased the initial compressive strength of cement mortar. Non-ureolytic calcium carbonate precipitation by AK13 using oyster shell-derived calcium ions was verified by energy-dispersive X-ray spectroscopy and X-ray diffraction analysis. Stereomicroscope and field emission scanning electron microscopy confirmed that oyster shell-derived calcium ions, along with soybean meal-solution, increased the bacterial survival and calcium carbonate precipitation inside mortar cracks. These data suggest the possibility of commercializing bacterial self-healing concrete with economical substitutes for culture medium, growth nutrient, and calcium sources.

Summary:

However, the high production cost of bio-concrete should be reduced for its complete commercialization. Three factors are required to produce bacterial self-healing concrete: carbonate-producing bacteria, nutrients for their growth and metabolism, and a calcium source to promote calcium carbonate precipitation. Oyster shells have been mixed with concrete as a building material to solve the environmental problems, but there are no precedents for its use in bio-concrete. The purpose of this study is to explore the possibility of using agricultural by-products and oyster shells as economical substitutes for nutrients for bacterial culture, metabolic activities, as well as calcium source which required to manufacture bio-concrete. D Comparison of the amount of calcium carbonate precipitated by AK13 with the calcium ions separated from the same mass of calcium lactate and oyster shell powder Measurement of mortar compressive strengths Calcium ions isolated from oyster shells were mixed into the mortar to measure the effect on the compressive strength, which is one of the most important properties of concrete. Nitrate-extracted OC is expected to be a cost-effective admixture to bio-concrete that compensates for the organic matter-induced decrease in compressive strength. Among the six agricultural by-products, soybean meal was most effective in germination and growth of AK13 spores, and thus was used as a nutrient to replace the yeast extract that has been used in decades of bio-concrete research. The increase in crack-sealing rate due to SN treatment was particularly evident in Mix and Spo mortars, which suggest an increase in the bacterial metabolic rates due to the addition of nutrients, which resulted in the increased MICP in the cracks of Mix and Spo mortars. Calcium ions extracted from oyster shells had a positive effect on the preservation of bacteria and crack-sealing.

Mentioned biomedical entities:

Genes: Mrp, Mrp antiporter, OC

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AK13 cells, Fig. S2). Higher spore, Fishers F, ISO, L. sphaericus, S. pasteurii, SM, SM-medium, Spo, Spores, b, b, b., bacterial cells, biofilm, by-product cultures, calcite-form calcium carbonate, cell, cell wall-bound calcium, cells, colonies, layer, lines, lysozyme, screening AK13 cells, seed culture, soybean, spore

Organisms: human, humans

Tissues: soybean pulp, specimens, tube

Pathways: N/A

Query Result 23/25

Microbial-induced calcium carbonate precipitation: an experimental toolbox for in situ and real time investigation of micro-scale pH evolution

Zehner, Jennifer; Ryne, Anja; Wentzel, Alexander; Sikorski, Pawel

RSC Advances

Year: 2020

PMID: 35517729

doi: 10.1039/d0ra03897k

Citations: 0

Score: 10.36

Keywords hits: concrete: 6, calcite: 29, precipitation: 46

Abstract:

Concrete is the second most consumed product by humans, after water. However, the production of conventional concrete causes more than 5% of anthropogenic CO₂ emissions and therefore there is a need for emission-reduced construction materials. One method to produce a solid, concrete-like construction material is microbial-induced calcium carbonate precipitation (MICP). To get a better understanding of MICP it is important to be able to follow local pH changes in dissolution and precipitation processes of CaCO₃. In this work we present a new method to study processes of MICP at the micro-scale in situ and in real time. We present two different methods to monitor the pH changes during the precipitation process of CaCO₃. In the first method, the average pH of small sample volumes is measured in real time, and pH changes are subsequently correlated with processes in the sample by comparing to optical microscope results. The second method is introduced to follow local pH changes at a grain scale in situ and in real time. Furthermore, local pH changes during the dissolution of CaCO₃ crystals are monitored. We demonstrate that these two methods are powerful tools to investigate the pH changes for both MICP precipitation and CaCO₃ dissolution for knowledge-based improvement of MICP-based material properties.

Summary:

1 Principle of global and local p H monitoring for precipitation: *S. pasteurii* dilution is added to the crystallization solution and phenol red in a 96-well plate. Additionally, it can be observed that the ACC and vaterite formed in Stage I and visible as small particles in the brightfield images, as well as local p H changes in the p H mapping, is starting to dissolve, and the nucleated crystals increase in size. During the first 5 min of the experiment, the area with higher p H is decreasing, while the p H of the entire sample increases. During the dissolution process, two phases can be observed. Once a uniform p H in the sample is established, the main observed process in local p H monitoring experiments is the increase of the p H of the entire sample. The p H profiles have been plotted along the dashed line shown in Fig. The closed conditions of the flow cell experiments lead to a faster p H increase and as a consequence faster precipitation of Ca CO₃. Furthermore, the influence of the surface morphology of the substrate on precipitation and morphology of precipitated crystals in MICP has been reported earlier.³⁵ This shows that precipitation depends on the surface chemistry of the substrate, and can contribute to the different time-points at which precipitation starts in global and local p H monitoring experiments.4.2 Dissolution of Ca CO₃ crystals In our dissolution experiments, we observed a slow dissolution of calcite crystals (see Fig.

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, Jack Bean urease, bacterial cells, cell, cells, earlier.³⁵, first.¹⁶, flow-cell, processes:267CaCO₃ +, system.14CaCO₃, wells

Organisms: N/A

Tissues: Falcon tube

Pathways: N/A

Query Result 24/25**Development of biological techniques to prevent corrosion of reinforcing steel bars**

Rasheed, Sajid; Khushnood, Rao Arslan; Raza, Ali; Ahmed, Sajjad; Kanwal, Maria

Heliyon

Year: 2024

PMID: 39323773

doi: 10.1016/j.heliyon.2024.e37966

Citations: 0

Score: 10.35

Keywords hits: concrete: 64, calcite: 23, precipitation: 3

Abstract:

The electrochemical corrosive processes compromise the passivity of reinforcing steel, potentially leading to structural integrity loss and, in extreme cases, concrete infrastructure failures. While bio-inspired concretes show promise in mitigating strength degradation and enabling self-healing of concrete flaws, their interaction with steel reinforcement remains underexplored. Thus, this investigation aimed to establish a protective strategy by fostering biofilm growth on rebar surfaces. To achieve this, *Bacillus subtilis* and *Escherichia coli* bacteria were utilized as biofilm-forming agents, aided by magnetic iron oxide and zeolite micro-nano particles. The study encompassed a thorough assessment of split tensile strength, corrosion resistance of bio-treated embedded steel bars, and a comprehensive biofilm characterization, along with a meticulous examination of the microstructure at the steel-concrete interface. The findings underscored a significant improvement in split tensile strength, demonstrating a remarkable 84.2% increase when bacterial species were combined with iron oxide nanoparticles, in contrast to the control specimens. Furthermore, the bio-treated bars exhibited an impressive corrosion inhibition potential of 78.5% relative to their unaltered counterparts. These outcomes are attributed to the discernible refinement of microstructural features surrounding the steel reinforcement and the heightened densification of the inter-transitional zone between steel and concrete.

Summary:

3. 9. Fig. 16. Fig. A corrosion rate of 16.3 was reported by Ghafari et al., 213.5 Conclusion It is concluded from this study that concrete microstructure refinement and biofilm around the steel bar led to the following results: Biofilm with an average thickness of 12 m was successfully developed on steel rebar to act as the passive protective layer against corrosion for *Bacillus subtilis* with iron oxide, least biofilm was developed by the *e. coli* without any carrier medium which was 7 m on average which is more than *P. aeruginosa* resulted in 6 m biofilm development previously by SV Oopath et al., 222. Biologically encapsulated steel-embedded concrete provided 78.5 % corrosion inhibition efficiency and 998 polarization resistance which is 159 % more than the control. The corrosion rate was reduced by 6.25 % due to the protective biofilm layer and refinement of the concrete microstructure *e. coli* without carrier medium showed the least corrosion resistance of 29.4 % owing to the thin protective layer. Split tensile strength was successfully improved by 84.2 % due to the roughness of biofilm and bacteria calcite precipitation in the interfacial bond zone, *e. coli* without carrier medium resulted in the least improvement of 19.4 %. Bond strength of steel-embedded concrete specimens was enhanced by 59.8 % at maximum when *Bacillus subtilis* with iron oxide was used as a protective layer around steel rebars, followed by the *E. coli* with iron oxide showed a 55.2 % improvement in bond strength than the control mix. 6 Limitation and recommendation This study is limited to normal bacterial strains which are not strained in any chemical environment.

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Biofilm, *E. coli*, *E. coli*, *E. coli*, EC, SEM, b) *Bacillus*, b) *E. coli*, b) EDX, b) *Escherichia coli*, b) SEM, bacterial cells, biofilm, biofilm layer, cathode, *e. coli*, electrolyte, layer, material, solid, specimens, *subtilis*Organisms: *Bacillus subtilis*, *E. coli*, *Escherichia coli*, SV Oopath et, *Bacillus subtilis*, human body

Tissues: VSM, layer, sieve, sieve no., specimens

Pathways: N/A

Query Result 25/25**Biomining of calcium carbonates and their engineered applications: a review**

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Keywords hits: concrete: 13, calcite: 33, precipitation: 32

Abstract:

Microbially induced calcium carbonate precipitation (MICCP) is a naturally occurring biological process in which microbes produce inorganic materials as part of their basic metabolic activities. This technology has been widely explored and promising with potential in various technical applications. In the present review, the detailed mechanism of production of calcium carbonate biomaterials by ureolytic bacteria has been discussed along with role of bacteria and the sectors where these biomaterials are being used. The applications of bacterially produced carbonate biomaterials for improving the durability of buildings, remediation of environment (water and soil), sequestration of atmospheric CO₂ filler material in rubbers and plastics etc. are discussed. The study also sheds light on benefits of bacterial biomaterials over traditional agents and also the issues that lie in the path of successful commercialization of the technology of microbially induced calcium carbonate precipitation from lab to field scale.

Summary:

Though, the precise role of bacteria and bacterial activities in carbonate crystallization remains unclear, they seem to fall in 3 categories: As per the first hypothesis, mineralization occurs as a byproduct of microbial metabolism As per the second hypothesis, carbonate nucleation takes place on the cell wall and Lastly, the third hypothesis involves role of extracellular macromolecules. Mainly four groups of microorganisms are seen to be involved in the process Photosynthetic organisms such as cyanobacteria and algae Sulphate reducing bacteria that are responsible for dissimilatory reduction of sulphates Organisms utilizing organic acids Organisms that are involved in the nitrogen cycle either ammonification of amino acids/ nitrate reduction/ hydrolysis of urea. Of all the above, Microbially induced calcium carbonate precipitation via urea hydrolysis is the simplest and most widely used method for precipitation of carbonates for several technical applications. MICCP via urea hydrolysis The precipitation of carbonates via urea hydrolysis by ureolytic bacteria is the most straightforward and most easily controlled mechanism of MICCP with potential to produce high amounts of carbonates in short period of time. These bacteria include *Bacillus pasteurii*, *Pseudomonas* sp., *Variovorax* sp., *Leuconostoc mesenteroides*, *Micrococcus* sp., *Bacillus subtilis*, *Deleya halophila*, *Halomonas eurihalina* and *Myxococcus xanthus*. Table 1 Reaction conditions reported in the literature for production of Ca CO₃ via urea hydrolysis. Purpose Urea Ca²⁺ Urease activity Reference Sr9 sequestration33325.45 Fujita et al., 2 Sr9 sequestration33.25.42 Warren et al., 21 Removal of Ca²⁺ from waste water1614.293 Hammes et al., 23 Removal of Ca²⁺ from waste water815.32 Hammes, 22 Stone remediation333125.11 Stocks-Fischer et al., 199933334.2.12 De Mynck et al., 211 Stone remediation6625.41 Bachmeier et al., 22 Portland cement remediation3335n/s Ramachandran, 21 Plugging of rock pores333.25n/s Gollapudi et al., 1995 Biocementation1515418 Whiffin, 24 Sand column cementation33325.65 Achal et al., 29a Van Paassen, 29 Soil cementation11n/s Cheng et al., 213 Biodeposition33334n/s De Belie and De Mynck, 28 Carbonate precipitation66625n/s Okwadha and Li, 21 Polymorphism of carbonate crystals Studies have reported that bacterial induced calcium carbonate precipitation results in the production of different phases of Ca CO₃. , bformula id=E159 Sr²⁺+OH+HCO₃=Sr CO₃+H₂O These cations and radionuclides integrate into the calcite structure via substitution of calcium ions in the microenvironment of the mineral precipitate, forming low - strontium carbonate minerals which have very low solubility. Recently Achal et al. reported the positive potential of bacterial calcite in crack remediation by *Bacillus* sp. Dhami et al. successfully improved the durability of these low energy building materials by application of ureolytic *Bacillus* sp.

Mentioned biomedical entities:

Genes: Arp, CSL, IAP, MR1, PU, RHA

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: BCM, Ehrlich, Mitchell, SEM, algae, b, bacterial cells, cell, cells, column cementation33325.65Achal, limestone specimens, microbes, solid, specimens, ureolytic *S. pasteurii* cultures, wheat strawOrganisms: *Bacillus subtilis*, Bang, Ferris, bovine CA, donors, human, humans, kidney stones, stone, teeth

Tissues: globe, layer, specimens, stone

Pathways: N/A